# Evidentiary reasoning within the Analytical Hierarchy Process: Use of graph entropy to rank decision elements

unpublished working paper

GitHub: hamlett-neil-ur/ahp-graph-entropy

Neil A. Hamlett, D.Sc., MBA

neil.hamlett@uncertainty-research.science neil-hamlett-strategic-quant

hamlett-neil-ur.github.io ORCID: 0000-0001-8278-0087

May 28, 2026

## Abstract

Decision makers often require insight into the relative strength of the evidence on which their choices hinge. The Analytic Hierarchy Process (AHP) contains innate mechanisms leading to the rigorous quantification of evidentiary elements' relative strength. Comparative judgements produce lists of edges for a complete, directed, acyclic graph, often called a tournament graph. The resulting Intensities of Importance provide tournament-graph edge weights, leading to a novel Laplacian matrix. Relative-entropy calculations based on Laplacians for induced subgraphs quantify the relative evidentiary strength associated with the decisional evidence ommitted from them. A previously published case study related to a fuel-cell design demonstrates the approach. The graph-entropy method provides greater strength-of-contribution contrast than sensitivity analysis from the case study.

---

## Introduction.

Decision-making nearly always occurs under conditions of incomplete or imperfect information. No less than Nobel-prize economist Frederick von Hayek recognized this:

> "…I prefer true but imperfect knowledge, even if it leaves much undetermined and unpredictable, to a *pretence of exact knowledge that is likely to be false*…" (emphasis added) (von Hayek (1974), cited in Kay & King (2020)).

The treatment of uncertainty in quantitative decision making represents a formidable research challenge confronting multiple facets. We conventionally decompose uncertainty into *epistemic* and *aleatory* (or *stochastic*) components (Salicone & Prioli (2018)). Sound evidentiary reasoning must account for the consequences of both.

*The present study focuses primarily on the epistemic.* Hamlett (2019, 2025) argues for the primacy of resolving epistemic uncertainty in business decision making. Exemplified by Zellweger & Zenger (2023); Rindova & Martins (2024); Townsend, et al (2025); Ehrig, et al (2026), this perspective holds sway in the entrepreneurial and strategic decision-making communities. Jackson (2019) corroborates from a philosophy-of-science point of view.

### Motivation: How strong is our evidence?

Ehrig, et al (2026) argue that entrepreneurial decision making in particular inescapably requires "…logic and reasoning with the aid of *very small samples of data*" (emphasis added). Circumstances demanding this sort of reasoning arise in numerous contexts beyond new-business formation (Martin & Olsberg (2007); Eisenmann (2013); Swanson (2024)). Fundamentally, entrepreneurial ventures amount to *epistemic* explorations (Camuffo, et al (2024)). Howard & Abbas (2016) devoted considerable attention to the adverse effects of imperfect information — an endemic condition within entrepreneurship — when trying to select optimum scenarios.

Nontrivial challenges inevitably arise even when considerable amounts of statistically coherent, phenomenologically illuminating data exist. Cover & Thomas

(2006) derive an information-theoretic *data-processing inequality*: "...no processing of $Y$ ... can increase the information that $Y$ contains about $X$." Corroborating from a statistical-learning perspective, Hastie, Tibshirani, Friedman (2009) define *irreducible variance*. Kay & King (2020) warn of pervasive *nonstationarity* in real-world problems: Statistical-learning training data based on past experience might not represent future conditions very well. More generally, Townsend, et al (2025) articulate the case for "...computationally irreducible barriers to the predictive capabilities and task performance advantages of AI systems..." for many tasks.

Extending Dempster's (1967) seminal concept, Shafer (1976) offered a rigorous approach to quantifying epistemic uncertainty in evidentiary reasoning. More-recently, Shenoy (2023) applied Shafer's reasoning to quantify evidentiary incompleteness in Bayesian networks, a causal-inference methodology commonly employed within the entrepreneurial decision-making community (Camuffo, et al (2024); Ehrig, et al (2026)).

Hué, et al (2026) use a stochastic-detection area-under-the-curve (AuC) approach (*cf.* Fawcett (2026)) to ascertain evidentiary elements' relative strengths. Resembling stepwise regression (Olive (2017)), their approach also exhibits conceptual parallels to Shafer (1976). Cox (2026a,b) offers an emerging approach to uncertainty characterization based on sheaf theory (Rosiak (2022)).

## Why the Analytical Hierarchy Process (AHP)?

AHP provides an axiomatic framework (Saaty (1986)) for "...logic and reasoning with the aid of very small samples of data" (Ehrig, et al (2026)). It is a multi-criterion decision methodology (MCDM) enjoying widespread applicability to group decision making (Saaty (1994)), risk analysis (e.g., Saaty (1987); Vaidya & Sushil (2006); Linkov, et al (2006)), and other problems. AHP is often applied to situations where substantial amounts of statistically coherent measurements are not immediately available to support more-empirical approaches.

The AHP methodology begins with identifying a plausible *evidentiary span*: The range of measurable factors — *elements* in the vernacular of Saaty (1986, 1994) — believed to differentiate within the range of scenarios under consideration. A *hierarchy model* represents this as a tree graph. The responsible individual decision maker or panel of experts then performs *comparative judgements*: "...the numerical representation of a relationship between two elements that share a common parent" in the tree graph (Saaty (1994)). A case study subsequently demonstrates AHP's use to rank plausible scenarios.

Not strictly a stand-alone method, AHP can be fused with other approaches. Bayesian techniques allow for subsequent revision of the comparative judgements as stronger empirical evidence emerges (Merrick, van Dorp, Singh (2005); Li, H., et al (2025)). Li, J. et al (2025) combined back-propagation neural networks with the AHP to design shields for neutron generators. Xiao, et al (2025) explored simulation-optimization approaches to circumvent altogether the elicitation of comparative judgements from experts. Lukinskiy, Lukinskiy, Bazhina (2025); Sáenz-Royo & Chiclana (2026) studied methods to improve the rubustness of expert inputs into comparative judgements. Syed, Shabarchin, Lawryshyn (2020) describe AHP's use in eliciting prior probabilities for Bayesian reliability analysis.

## Evidentiary-strength analysis using the AHP.

That the *comparative judgements for an AHP hierarchy model define a complete, directed, acyclic graph* (DAG) provides the seminal insight underlying this investigation. Saaty (1986) recognized this: "Partially ordered sets with a finite number of elements can be conveniently represented by a directed graph." Nishizawa (1999) obliquely considered graphical methods for AHP consistency. The decision-theory literature however contains scant exploration of AHP's graphical characteristics and opportunities arising therefrom.

Complete DAGs belong to a special class of graphs called *tournament graphs* (Moon (2015)). In general, "the entropy of a graph is interpreted *as its structural information content* and serves as a complexity measure" (emphasis added) (Dehmer & Mowshowitz (2011); Mowshowitz & Dehmer (2012)). Structural information content bears direct relevence to the present concern of evidentiary reasoning. Brown, et al (2020) studied the particular characteristics of tournament graphs.

**What questions do entropies of AHP-based tournament graph allow us to answer?** Endogenous to the AHP formulation itself, *relative entropy* quantifies the marginal informational contribution by a subset of elements within the evidentiary span. This addresses the objective pursued by Hué, et al (2026). Graph-entropy measures however do not require extensive empirical data spanning of both positive and negative cases, as with Hué's approach. Congruent with the AHP methodology explored here, Jiroušek, Kratochvíl, Shenoy (2023) use entropy of *belief-function graphical models* to decompose their elements' relative evidentiary strength.

**By what mechanism do we infer the differential information between hierarchically-coupled AHP hierarchy**

**models?** Relative entropy provides a well-understood approach (Cover & Thomas (2006)). Relative entropy for graphs has been well-researched (Wong & You (1985); Liu, et al (2022); Guanlei, Xiaogang, Xiaotong (2025). Brown, et al (2020) explored entropy differences between same-order tournament graphs with differing directional structures. In our case, the relative entropy of the graph associated with the overall evidentiary span with respect to a constituent induced subgraph quantifies the information lost by ommission of elements from the latter.

**What innovation is required to define a methodology for AHP-based evidentiary-strength analysis?** First, we must formulate an entropy definition for tournament graph with *weighted* edges. Brown, *et al* (2020) analyzed graphs with unweighted (uniformly unit-weighted) edges. Chen, et al (2015), Kuo (2022), and Liu, et al (2022) formulate entropies for regular (non-directional) graph comprised of weighted edges. The intensities of importance from the AHP comparative judgements provide the weights used here.

Second, relative entropies of hierarchically coupled hierarchy models must be coupled. The *Hierarchy* term in AHP refers to a hierarchic deconstruction of decisions into contituent elements. The AHP method produces a set of decision-element comparative judgements — represented by a tournament graph — for each hierarchy model. (Cover & Thomas , 2006, §2.5) describe *chain rules* for entropy, relative entropy, and mutual information. Such mechanisms offer a rigorous approach to link hierarchically adjacent tournament graph.

**Why is "heavy" graph-entropy machinery needed?** We might intiutively expect that individual comparative judgements suffice to inform us about the most-influential elements. The AHP's machinery however accounts for "mutual coupling" between elements: "...many of the concepts derived for hierarchies also relate to systems with feedback" (Saaty (1986)). That comparative judgements define a graph — a network — provides the basis of its power. Our graph-entropy machinery accounts for the *aggegate effect* of all the evidence informing a decision or risk assessment. Individual importance intensities fail to provide this.

## *Decision Quality* as a check on the completeness of evidentiary reasoning.

*Decision Quality* (DQ) represents a prominent applied decision-analysis framework that emphasizes evidentiary reasoning (Spetzler, Winter, Meyer (2016)). Philosophically congruentment with design-of-experiements practice (Cox & Reid (2000)), DQ provides a *de facto* standard of practice for the Society of Decision Professionals (SDP). SDP is a community of applied decision analysts and scientists. Introducing DQ here — independent of the mechanics of AHP — reinforces our logical completeness.

The pattern of reasoning underlying Saaty's (1986) AHP methodology loosely maps to the DQ framework. Figure 1 depicts the DQ framework, The DQ frameworks contains six *links*:

- Establishing an *Appropriate Frame* specifies the scope of the decision: What specifically is to be decided. Thoughtful framing combines prescriptive with prospcriptive judgements to provide analytical clarity. Saaty (1994) describes this as "...the most creative part of decision making that has a significant effect on the outcome..."

- By definition, the act of deciding involves selection of one or more options from a discrete, precisely defined set of *Creative Alternatives*.

- *Relevant and Reliable Information* constitute the evidence weighed in selecting the best alternative.

- *Clear Values and Tradeoffs* specify the criteria by which the best alternative is identified. They specify the decision makers' values, and justify the optimality of the best alternative.

- *Sound Reasoning* defines the logic employed in identifying the best alternative. Creighton (1909) cautioned that "...it is not possible to lay down definite rules for the guidance of thinking in every case." The reasoning approach must therefore fit the circumstances resulting from the links above. AHP is itself a distinct reasoning approach. Injudiciously applied, it does not however guarantee satisfaction of the von Neumann-Morgernstern axioms (Kochenderfer , 2015, p. 58).

- *Commitment to Action* represents the follow-through in implementing or adopting the selected alternative. Absent a commitment to adopt or act upon the best alternative resulting from the links above, a decision arguably has not occurred. The analysis was academic.

Now, use of AHP by no means suffices to achieve DQ. Thoughtful application of AHP does adress a subset of DQ's links. Featuring AHP as an approach to DQ does not however define our objective. Here we explore AHP's ability to highlight sets of decisional values and spans of reliable information influence the results of sound reasoning.

Figure 1: Decision-Quality framework. (From Society of Decision Professionals (DQ))


## Course of investigation.

We begin with a brief illustration of AHP methodology. A *real-world* case study concisely demonstrates its application to preference-ranking a set of scenarios. The actual ranking lies beyond our specific interest, but is included for completeness. The DAGs resulting from the comparative judgements provide our point of departure into a novel methodology.

We then calculate the entropies for our comparative-judgement DAGs associated with distinct hierarchy models. This extends Brown, et al (2020) through adoption of a Laplacian structure that includes edge weights. Our AHP-graph Laplacian fuses the structure of a conventional unit-edge-weighted DAG Laplacian with comparative-judgement importance intensities, which provide the edge weights. This incorporates information about degree of influence in the interconnected evidentiary framework.

Finally, we calculate entropies and relative entropies from our AHP-based DAG structure comprised of multiple, hierarchically coupled hierarchy models. We interpret the results for our real-world case study. Approaches analogous to those by Jiroušek, Kratochvíl, Shenoy (2023) guide our thinking. We also explore the relationship between importance intensities and relative entroy. This quantifies relative strengths of contribution by subsets of evidentiary elements.

# AHP case study .

Bu, et al (2026) employed AHP in making tradeoff decisions during the design of Proton Exchange Membrane Fuel Cells (PEMFCs), which "...efficiently convert chemical energy from fuels into electricity without causing polution...". Bu, et al (2026) describe device-physics technical details that are tangential to our purposes. The AHP mechanics primarily interests us.

Bu, et al (2026) particularly suits the purposes of this investigation for four reasons.

1. *It demonstrates a real-world application.* Introductory AHP presentations often employ contrived cases. A real-world case study provides a more-informative and -convincing demonstration.

2. *This case is recent.* Given that the AHP literature spans decades, a reader might succomb to the temptation to conclude that the method is anochronistic. Our case here demonstrates the application of AHP to a contemporary design problem for an economically essential application.

3. *It strikes a balance between tractability and complexity.* Many pedantic demonstrations are borderline-trivial, constructed to illustrate AHP mechanics. The present investigation benefits from a moderately greater degree of complexity. The selected case provides the opportunity to demon-

strate the use of relative entropy to couple hierarchically adjacent hierarchy models.

4. *The case emphasizes sensitivity analysis*. This aligns well with the focus of the present investigation. Although Bu, et al (2026) ultimately arrived at an optimum design configuration, their extensive sensitivity analysis provides a baseline against which to compare the proposed graph-entropy approach.

The following offers neither criticism nor endorsement of Bu, et al (2026). Their work merely serves to illustrate AHP mechanics.

We strive in the following to select AHP terminology that remains close to Saaty (1986, 1994). Variations in terminology occur througout the AHP-application literature. For example, the quantity Saaty (1986) defines as a *priority vector* is frequently referred to as a "weight vector" or "weights". We employ *priority vector* here.

## Specification of the evidentiary span.

Figure 2 reproduces the *hierarchy model* employed by Bu, et al (2026) for their fuel-cell design. Visual, tree-graph depiction of the decision-element hierarchy aids in understanding by human decision-makers. Idiosyncratic style notwithstanding, Figure 2 follows the ubiquitous form patterned after Saaty (1986, 1994), and appearing throughout the MCDM literature. Structural elements of Figure 2 map to key DQ links from Figure 1.

- The rectangular box "PEMFC performance" at the top represents the *Goal or Focus*, in Saaty's (1986) terms. A clear goal or focus results from thoughtful *framing*, in DQ terms.

- The curve-cornered boxes represent *Criteria*. Corresponding to DQ's *values*, the these define the span of measurable criteria believed to materially distinguish between alternatives. For this case, they represent device physical-operating parameters *current density* $I$, *pressure drop* $\Delta P$, and *Oxygen-uniformity index* $UI$. These physical-operating parameters decisively influence the objective.

- The heptagonal shapes at the bottom depict decomposed criteria. Saaty (1986) describes "...intermediate set of higher criteria that can both be decomposed into these attributes and are themselves decompositions of the higher level criteria or subcriteria...". In this case, each of three physical-device geometric specifications — angle $\theta$, width $W$, and height $H$ — is believed to influence the operating parameters above.

Figure 2: Hierarchy model used for Proton Exchange Membrane Fuel Cell (PEMFC) case study. (From Bu, et al (2026), Figure 5)

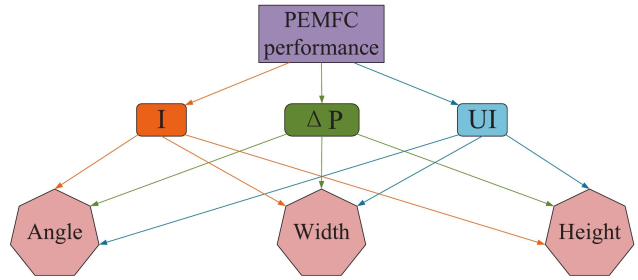


Figure 2 omits the *discrete alternatives* from which the decision maker selects. Hierarchy-model depictions often include the alternatives when the set is small. Bu, et al (2026) however computationally simulate device performance based on 36 distinct combinations of geometric parameters to construct distinct alternatives. The clutter arising from depicting each scenario in Figure 2 would detract from its clarity. Using AHP to analyze the risk of a large porfolio of distinct assets exemplifies another scenario in which the alterantives do not explicitly appear in the hierarchy.

The criterion hierarchy in Figure 2 is also idiosyncratic for another reason: Each of the geometric-specification sub-criteria influences the each operating-parameter criteria. Typically, constituent subcriteria are distinct to their associated criterion. Their representations appear as "clean" tree graph. This excursion from common practice does not invalidate AHPs application, here. One might alternatively consider the *Analytic Network Process* (ANP) for scenarios like this (Saaty & Vargas (2013); Chen, et al (2019)).

Finally, Figure 2 includes an uncommon feature that clarifies interpretation: It shows edge directivity. Always directed tree graphs, AHP hiearchy hierarchy-models tend to omit edge direction. Its appearance here reinforces our own graphical interpretation of AHP.

## Comparative judgements about relative importance of criteria.

AHP imbues values and tradeoffs into decision-making through comparative judgements: "...a matrix to carry out *pairwise comparisons* of the relative importance of the elements [criteria]..." (emphasis added) (Saaty (1986)). *Pairwise* is the operative term, here. Practically, one assembles a list of unique, unordered element pairs for each hierarchy model. If the hierarchy model for a given objective (or crietrion) spans $n$ elements, then $\binom{n}{2}$ element pairs must be considered.

Figure 2 contains four hierarchic models, each based on three constituent elements. Consequently, $\binom{3}{2} = 3$ element pairs must be considered for each. Each element pair corresponds to a directed edge for the tournament graph by which the hierarchy model is represented.

Table 1 contains the comparative judgements assigned by Bu, et al (2026) for each hierarchy-model element pair depicted in Figure 2. We defer construction of coresponding matrices to the next subsection. Explicit depiction of the comparative judgements here contributes to the clarity of present course of reasoning. Moreover, the directed-graph edge list in Table 1 provides a key of departure for subsequent extension of the AHP methodology.

The "O-C" model corresponds to the highest level of the hierarchy model in Figure 2. It represents the contribution by each of the device operating parameters $I$, $\Delta P$, and $UI$ to overall performance of the PEMFC device. The constituent hierarchy models account for the belief that geometric-design parameters angle $\theta$, height $H$, and width $W$ influence the operating parameters.

Now, (Saaty , 1986, Table 1) and (Saaty , 1994, Table 1) specify a *fundamental scale* containing arbitrary, subjective heuristics to represent the *intensity of importance* between criterion pairs. Bu, et al (2026) employ this scale (their Table 5), as do many other AHP applications appearing in the literature. Saaty (1986), the most-rigorous description of the AHP methodology, contains no explicit prescription for the necessity of these values.

Since Saaty (1986) speaks to preferential ordering, alternative, more-rigorous importance-intensity assignments can be considered. Application of the personal-probability indifference principle arises as one option (Savage (1972); de Finetti (1974)). Hubbard & Seiersen (2023) — addressing the specific domain of cybersecurity — describe more-emprical approaches to *measurement*. Application of such alternatives in lieu of Saaty's fundamental scale deflects a common critique of AHP's logical rigor.

The construction of Table 1 anticipates its use in defining a directed graph in the sequel. The AHP literature — Saaty (1986, 1994, 2003), in particular — presscribes nothing about the ordering of element pairs in comparative judgement. Construction of Table 1 employed the following convention.

- Populate the "intensity of importance" column with the value greater than unity.
- Assign to the "reference element" column the most-preferred element, associated with the prefered intensity-of-importance value. This element subsequently corresponds to the head of the directed graph edge connecting the two elements.

Table 1: Comparative judgements for each hierarchy-model criterion pair. (After Bu, et al (2026), Tables 8, 9, 10, and 11)

| hierarchy model | reference element | comparative element | intensity of importance |
|---|---|---|---|
| O-C | $I$ | $\Delta P$ | 7 |
| O-C | $I$ | $UI$ | 4 |
| O-C | $UI$ | $\Delta P$ | 3 |
| C-$\Delta P$ | $H$ | $W$ | 7 |
| C-$\Delta P$ | $H$ | $\theta$ | 5 |
| C-$\Delta P$ | $\theta$ | $H$ | 3 |
| C-$UI$ | $\theta$ | $H$ | 5 |
| C-$UI$ | $W$ | $H$ | 5 |
| C-$UI$ | $W$ | $\theta$ | 3 |
| C-$I$ | $W$ | $\theta$ | 5 |
| C-$I$ | $W$ | $H$ | 3 |
| C-$I$ | $H$ | $\theta$ | 2 |

- Populate the "comparative element" column the less-preferred element. This element corresponds to the tail of the directed-graph edge.

This convention imbues two conveniences into the procedure. First, the structure of the table endogenously contains the directive structure of the graph. This simplifies the algorithmic logic used to construct the resultant tournament graph. Second, it inclines us to think of importance intensities in terms as personal probabilities (Savage (1972); de Finetti (1974)) in lieu of the arguably problematic "fundamental scale".

## Transformation of comparative judments into *priority vectors.*

AHP *priority vectors* provide the key quantitative mechanism for evidentiary aggregation in MCDM applications. Mechanically, this is accomplished by constructiong a *reciprocal matrix* from the comparative judgements for each hierarchy model. A reciprocal matrix is a Hermetian matrix with unit-valued diagonals and intensity-importance values assigned to off-diagonal elements. For the O-C model in Table 1, this appears as

$$A_{\text{O-C}} = \begin{bmatrix} 1 & a_{I,\Delta P} & a_{I,UI} \\ a_{\Delta P,I} & 1 & a_{\Delta P,UI} \\ a_{UI,I} & a_{UI,\Delta P} & 1 \end{bmatrix} \tag{1a}$$

$$= \begin{bmatrix} 1 & 7 & 4 \\ 1/7 & 1 & 1/3 \\ 1/4 & 3 & 1 \end{bmatrix}. \tag{1b}$$

Equation (1a) follows from Table 1 and

$$\begin{matrix} a_{j,i} = a_{i,j}^{-1} \ \forall\, i \neq j \\ a_{i,i} = 1 \end{matrix} . \tag{1c}$$

If

$$A_{O-C}X_{O-C} = X_{O-C}\boldsymbol{\Lambda}_{O-C}, \tag{2}$$

represents the eigenvector/eigenvalue decomposition of reciprocal matrix $A_{O-C}$, the resulting priority vector $\omega_{O-C}$ becomes the Perron (or principal) eigenvector (Saaty (2003)):

$$[\omega_{O-C}]_i = \frac{[X_{O-C}]_{i,\text{argmax}(\boldsymbol{\Lambda}_{O-c})}}{\sum_i [X_{O-C}]_{i,\text{argmax}(\boldsymbol{\Lambda}_{O-c})}}. \tag{3}$$

Table 2a contains the reciprocal matrix $A_{O-C}$ and the priority vector $\omega_{O-C}$ resulting from the comparative judgements (Table 1) for the PEMFC-performance hierarcy model. The results coincide — within roundoff precision — with those reported by Bu, et al (2026) Table 12.

Table 2a: Reciprocal matrix and priority vector resulting from Table-1 hierarchy model $A_{O-C}$.

| reference element | $I$ | $UI$ | $\Delta P$ | $\omega_i$ |
|---|---|---|---|---|
| $I$ | 1 | 4 | 7 | 0.705 |
| $UI$ | 1/4 | 1 | 3 | 0.211 |
| $\Delta P$ | 1/7 | 1/3 | 1 | 0.084 |

Table 2b contains the same information for the constituent hierarchy models $C-I$, $C-UI$, and $C-\Delta P$. The results do not coincide as closely with those from Bu, et al (2026) Table 12, which are not reported with high degrees of numerical precision. The python *scipy* library (Virtanen, et al (2020)) produced the $\omega_i$ values in Table 2b. Bu, et al (2026) do not report how their Table-12 calculations were performed. Lacking these details, exact reproduction proves difficult.

Table 2b: Reciprocal matrices and priority vectors based Table-1 hierarchy models $A_{C-\Delta P}$, $A_{C-I}$, $A_{C-UI}$.

| hierarchy model | element | $\theta$ | $W$ | $H$ | $\omega_i$ |
|---|---|---|---|---|---|
| $C-I$ | $H$ | 2 | 1/3 | 1 | 0.230 |
| $C-I$ | $W$ | 5 | 1 | 3 | 0.648 |
| $C-I$ | $\theta$ | 1 | 1/5 | 1/2 | 0.1220 |
| $C-UI$ | $H$ | 1/5 | 1/5 | 1 | 0.0856 |
| $C-UI$ | $W$ | 3 | 1 | 5 | 0.618 |
| $C-UI$ | $\theta$ | 1 | 1/3 | 5 | 0.230 |
| $C-\Delta P$ | $H$ | 5 | 7 | 1 | 0.740 |
| $C-\Delta P$ | $W$ | 1/2 | 1 | 1/7 | 0.0938 |
| $C-\Delta P$ | $\theta$ | 1 | 2 | 1/5 | 0.1666 |

The mechanics illustrated by equations (1a, 2, and 3) are completely general. That $\omega_{O-C}$ is a projection onto the unit simplex $\Delta^3$ where

$$\Delta^n = \left\{ x \in \mathbb{R}^n \mid \sum_i x_i = 1 \ \wedge x_i \geq 0 \, \forall i \right\}$$

is highly significant. All AHP calculation results are constrained to the interval $[0, 1]$. This inlcudes both AHP-calculation results and standardization of inputs demonstrated in a subsequent subsection.

## On inconsistencies in comparative judgements.

Glenn Shafer once opined, "...probability is not really about numbers; it is about the structure of reasoning..." (quoted in Pearl (1988), p. 15). The AHP methodology contains mechanisms to identify inconsistencies and contradictions in the structure of reasoning (Saaty (2003); Pant, et al (2022)). Here, we consider the types of inconsistencies in comparative judgements from which concerns might arise. We then review *consistency indices*, AHP's mechanism to quantify the degree of inconsistency. Finally, we consider alternative graph-theoretic approaches to mitigate inconsistencies in reasoning.

### Types of inconsistencies comparative judgements.

Saaty (1986, 2003) expressed concern about four types of inconsistencies that might adversely affect decision making aided by the AHP. Such opportunities particularly arise in comparative judgements. Saaty (2003) seemingly suggested that accomodation of logical inconsistency in the AHP method might be more a feature than a bug: "...a modicum of inconsistency may be considered as a good thing, and forced consistency without knowledge of the precise values as an undesirable compulsion."

The DQ framework strives for sound reasoning. Pant, et al (2022) articulates the goal of "...a decision-making process with some constraints wherein the task is to obtain the maximum benefit from the available resources to get the best achievable results." The consequences to these goals arising from avoidable inconsistencies bear circumspectful consideration.

**Circular reasoning.** Expressing the first inconsistency scenario from Saaty (2003), equation (4)

$$\begin{aligned} &A \succ B, \\ &B \succ C, \text{ and} \\ &C \succ A \end{aligned} \tag{4}$$

exemplifies a preference ordering that contains circular reasoning. This obviously codifies a logical contradiction, violating the *completeness* and *transitivity* rational-preference constraints of the von-Neumann-Morgenstern axioms (Kochenderfer (2015)). Furthermore, it also violates one of de Finetti's (1974) principles of coherence: "...no limits on the probabilities that an individual may assign, except *they must not be in contradiction among themselves*" (emphasis added).

American mathematician William Thurston argued that "Mathematics is not about numbers, equations, computations, or algorithms: *It is about understanding*" (quoted in (Cook , 2018, p. 76); emphasis added). Applying computational machinery like AHP to a contradictory logical construct does not enhance understanding. It invites confusion. Alerting producers of contradictory comparative judgements and facilitating elimination of the circular-reasoning instances better serves the goals of sound reasoning (Spetzler, Winter, Meyer (2016)) and best-observable results (Pant, et al (2022)).

**Variances between panel members' comparative judgements.** MCDM methods often support group decisions. This requires eliciting comparative judgements from a range of experts. Panel members will generally provide differing ordering preferences and importance intensities. Based on Saaty (2003), Table 3 illustrates cardinal inconsistencies in ordering preferences from comparative judgements by a panel of five experts. Saaty's (2003) illustration omits importance-intensity estimates.

Aggegation of elicited probabilities from panels of experts has been explored extensively (Spetzler & Staël Von Holstein (1975); Lindley, Tversky, Brown (1979); Barons, Mascaro, Hanea (2022)). If each individual contributors' comparative judgements are free of circular reasoning — a condition satisfied in Table 3 — then established elicited-probability aggregation methods apply. Equation (1c) provides the mechainsm to apply elicitation-aggregation methods when ordering preferences differ.

Some challenges may accompany the resulting probability distributions. Cooke (2026) studied the practice and observed, "Methods based on the 'wisdom of crowds' that aggregate point forecasts of multiple experts are unlikely to yield good performance in such cases, regardless of the precise method by which judgments are weighted and aggregated." The "fundamental scale" ((Saaty , 1994, Table 1)) elicits such point estimates. Li, H., et al (2025) explored elicitation of comparative judgements as interval estimates to circumvent this difficulty.

**Longitudinal comparative-judgement inconsistencies.** Finally, Saaty (2003) expresses concern that "...people would be required to be like robots unable to change their minds with new evidence...". Nothing elsewhere in the literature defining the AHP methodology proscribes such standard Bayesian updating (Saaty (1986, 1987, 1994)). In fact, recent research explicitly adapts AHP to Bayesian patterns of reasoning (Merrick, van Dorp, Singh (2005); Altuzarra, Moreno-Jiménez, Salvador (2007); Li, H., et al (2025)).

**Transitive inconsistencies.** Saaty (1986) defines a criterion for transitive consistency

$$\begin{gathered} a_{i,j}\, a_{j,k} = a_{i,k} \\ \text{where } a_{i,j} = [A_{\mathfrak{H}}]_{i,j} \end{gathered} \tag{5}$$

for reciprocity matrix $A_{\mathfrak{H}}$ associated with hierarchy model $\mathfrak{H}$. This provides the basis for consistency indices. Now, (5) imposes a stronger constraint than the von Neumann-Morgenstern transitivity axiom (Kochenderfer , 2015, p. 58)

$$(A \succ B) \wedge (B \succ C) \Rightarrow A \succ C, \tag{6}$$

which only constrains transitivity of ordering preferences.

Saaty (1986) contains an extensive set of definitions, axioms, and theorems culminating in the demonstration that, if the eigenvector/eigenvalue decomposition of a *consistent* reciprocal matrix $A_{\mathfrak{H}}$ is

$$A_{\mathfrak{H}} X_{\mathfrak{H}} = X_{\mathfrak{H}} \Lambda_{\mathfrak{H}}, \tag{7a}$$

then

$$\lambda_{\mathfrak{H}} = n \tag{7b}$$

where $n$ denotes the order of $A_{\mathfrak{H}}$ and $\lambda_{\mathfrak{H}} = [\Lambda_{\mathfrak{H}}]_{\text{argmax}(\Lambda_{\mathfrak{H}})}$.

Transitive consistency bears important decision-theoretic consequences: "...when alternatives are scaled through paired comparisons, adding a new alternative can change the relative ranking of the old ones when the judgements are inconsistent or when several criteria are used" (Saaty (1986)).

## A consistency index for reciprocal matrices.

A consequence of (7b),

$$CI = \frac{\lambda_{\mathfrak{H}} - n}{n - 1} \tag{8}$$

provides a commonly used *Consistency Index* for a reciprocal matrix$A_{\mathfrak{H}}$ (Pant, et al (2022)). From (7b), one obtains $CI$ =0 for a perfectly consistent reciprocal matrix.

A *Consistency Ratio*

$$CR = \frac{CI}{RI} \tag{9}$$

Table 3: Illustration of disparate preference orderings from parwise judgments by a panel of five experts. (After Saaty (2003))

| ordering preference 1 | incidence preference 1 | ordering preference 2 | incidence preference 2 | incidence total |
|---|---|---|---|---|
| $A \succ B$ | 2 | $B \succ A$ | 3 | 5 |
| $B \succ C$ | 3 | $C \succ B$ | 2 | 5 |
| $A \succ C$ | 5 | $C \succ A$ | 0 | 5 |

Table 4: Consistency indices for hierarchy models in PEMFC-design case study.

| hierarchy model ($\mathfrak{H}$) | $\lambda_{\mathfrak{Y}}$ | $n_{\mathfrak{H}}$ | $CI_{\mathfrak{H}}$ | $CR_{\mathfrak{H}}$ |
|---|---|---|---|---|
| $O - C$ | 3.032 | 3 | 0.0162 | 0.033 |
| $C - I$ | 3.004 | 3 | 0.0018 | 0.004 |
| $C - UI$ | 3.136 | 3 | 0.0678 | 0.139 |
| $C - \Delta P$ | 3.014 | 3 | 0.0071 | 0.014 |

accompanies the consistency index. In (9), the denominator $RI$ results from calculating $CI$ for a large number of reciprocal matrices with randomly generated comparative judgements. Tabulated $RI$ values are commonly repeated, such as Table 6 in Bu, et al (2026). Reproducing the common result with higher numerical precision, Donhegan & Dodd (1991) reported $RI = 0.4887$ for reciprocal-matrix order $n = 3$.

Table 4 contains the consistency analysis for the hierarchy models on which Bu, et al (2026) was based. Pant, et al (2022) cited a threshold $CR_{\mathfrak{H}} \lesssim 0.1$, above which one "...questions the credibility of judgements." Comparative-judgement transitivity (5) provides the basis for creduilty, here.

By this criterion, the $CR_{C-UI}$ value in Table 4 raises questions about the consistency of reasoning in comparative judgements underlying the $C - UI$ hierarchy model. Not-reproduced results in Bu, et al (2026) reported a lower value for $CR_{C-UI}$, albeit much closer to the threshold in Pant, et al (2022).

## Entropy of edge-weighted, directed acyclic graph.

## Data availability.

This GitHub repository ahp-graph-entropy contains the data and the python and Neo4j logic on which this paper is based.

## References

Abbas, A. (2026). "Is It Approximate? Or Is It A Fundamentally Flawed Decision-Making Method?" Society of Decision Professionals/INFORMS Decision Analysis Society webinar, May 13, 2026. May 2026 Webinar.

Altuzarra, A., Moreno-Jiménez, J. M., Salvador, M. (2007). "A Bayesian priorization procedure for AHP-group decision making." *European Jouranl of Operations Research*: 182(1)367-382. https://doi.org/10.1016/j.ejor.2006.07.025.

Barons, M. J., Mascaro, S., Hanea, A. M. (2022). "Balancing the Elicitation Burden and the Richness of Expert Input When Quantifying Discrete Bayesian Networks." *Risk Analysis*: 42(6):1196-1234. https://doi.org/10.1111/risa.13772.

Brown, D. E., *et al* (2020). "Entropy of tournament digraph." *Linear Algebra and its Applications*: 584:371-393. https://doi.org/10.1016/j.laa.2019.09.026.

Bu, W., *et al* (2026). "Multi-criteria evaluation of symmetric tesla-like valve flow channel geometry on proton exchange membrane fuel cell performance using analytic hierarchy process-entropy weight method." *Energy Conversion and Management*: 350:120952. https://doi.org/10.1016/j.enconman.2025.120952.

Camuffo, A., Gambardella, A., Pignataro, A. (2024). "Theory-driven strategic-management decisions." *Strategy Science* 9(4):297-514. https://doi.org/10.1287/stsc.2024.0173.

Caporale, D. & Rinaldi, A. (2025). "The application of analytical hierarchy process to assess adaptation strategies for flood and landslides risks: A case study of a multi-risk area community." *Environmental Science & Policy*: 163:103959. https://doi.org/10.1016/j.envsci.2024.103959.

Chen, Z., *et al* (2015). "Entropy of weighted graph with Randić weights." *Entropy*: 17:3710-3723. https://doi.org/10.3390/e17063710.

Chen, Y., et al (2019). "Analytic network process: Academic insights and perspectives analysis." *Journal of Cleaner Production*: 235:1276-1294. https://doi.org/10.1016/j.jclepro.2019.07.016.

Cook, M. (2018). *Mathematicians: An Outer View of the Inner World*. Providence, RI: American Mathematical Society. ISBN: 9781470448387.

Cooke, R. M. (2026). "Wisdom/Madness of Crowds and Perils of Point Forecasts." *Decision Analysis*: article in advance. https://doi.org/10.1287/deca.2025.0437.

Cover, T. M. & Thomas, J. A. (2006). *Elements of Information Theory*. Second edition. New York: Wiley. ISBN: 9780471748823.

Cox, D. R. & Reid, N. (2000). *The Theory of the Design of Experiments*. UK: Routledge. ISBN: 9781584881957.

Cox Jr., L. A. (2026a). "Integrating fragmented risk knowledge: Sheaf theory for risk analysts." *Risk Analysis*: 46(3):e70206. https://doi.org/10.1111/risa.70206.

Cox Jr., L. A. (2026b). "Reducing risk misinformation and miscommunication: A sheaf-theoretic perspective." *Risk Analysis*: 46(4):e70224. https://doi.org/10.1111/risa.70224.

Creighton, J. E. (1909). *An Introductory Logic*. London: MacMillan. LCCN: 33034164.

de Finetti, B. (1974). *Theory of Probability*. Chichester, UK: Wiley. ISBN: 9781119286387.

Dehmer, M. & Mowshowitz, A. (2011). "A history of graph entropy measures." Information Sciences: 181(1):57-78. https://doi.org/10.1016/j.ins.2010.08.041.

Dempster, A. (1967): "Upper and lower probabilities induced by a multivalued mapping." *Annals of Mathematical Statistics*: 38(2):235-339. https://doi.org/10.1214/aoms/1177698950.

Donegan, H. A. & Dodd, F. J. (1991). "A note on Saaty's random indices." *Mathematical and Computer Modelling*: 15(10)135-137. https://doi.org/10.1016/0895-7177(91)90098-R.

Ehrig, T., Felin, T., Zenger, T. (2026). "Causal Reasoning and the Scientific Entrepreneur: Beyond Bayes." *Bayesian Entrepreneurship*. Cambridge, MA: The MIT Press. ISBN: 9780262052160

Eisenmann, T. (2013, January). "Entrepreneurship: A Working Definition." *Harvard Business Review*. https://hbr.org/2013/01/what-is-entrepreneurship.

Fawcett, T. (2006). "An introduction to ROC analysis." *Pattern Recognition Letters*: 27(8):861-874. https://doi.org/10.1016/j.patrec.2005.10.010.

Guanlei, X., Siogang, X., Xiaotong, W. (2025). "Relative entropy based uncertainty principles for graph signals. *Signal Processing*: 227:109708. https://doi.org/10.1016/j.sigpro.2024.109708.

Hagberg, A. A., Schult D. A, Swart, P. J. (2008). "Exploring network structure, dynameics, and function using NetworkX." *Proceedings of the 7th Python in Science Conference (Scipy2008)*. Gäel Varoquaux, Travis Vaught, and Jarrod Millman (Eds), (Pasadena, CA USA), pp. 11–15, Aug 2008. https://doi.org/10.25080/TCWV9851.

Hamlett, N. A. (2019). "Competitive advantage based on information." unpublished working paper. https://drive.google.com/file/d/1Cq7oM6BYrGWxmjf1bdhrdrORZnKBvLu4/view.

Hamlett, N. A. (2025). "A theory and methodology for AI-based digital transformation." *Achieving Digital Transformation through Analytics and AI*, J. Liebowitz (ed.). Singapore: World Scientific. https://doi.org/10.1142/9789811296475_0002.

Hastie, T., Tibshirina, R., Friedman, J. (2009). *The Elements of Statistical Learning*. Springer Nature. https://doi.org/10.1007/978-0-387-84858-7.

Howard, R. A. & Abbas, A. E (2016). *Foundations of Decision Analysis*. New York: Pearson. ISBN: P200000003532.

Hubbard, D. W. & Seiersen, R. (2023). *How to Measure Anything in CyberSecurity Risk*. New York: Wiley. ISBN: 9781119892335.

Hué, S. et al (2026). "Measuring the driving forces of predictive performance: Application to credit scoring." *Management Science*: online first. https://doi.org/10.1287/mnsc.2023.02025.

Jackson, M. C. (2019). *Critical Systems Thinking and the Management of Complexity*. New York: Wiley. ISBN: 9781119118398.

Jiroušek, R., Kratochvíl, V., Shenoy, P. P. (2023). "Computing the decomposable entropy of belief-function graphical models." *International Journal of Approximate Reasoning*: 161:108984. https://doi.org/10.1016/j.ijar.2023.108984.

Kay, John & King, Mervyn (2020). *Radical Uncertainty*, New York: Norton. ISBN: 9781324004776.

Kochenderfer, M. J. (2015). *Decision Making Under Uncertainty: Theory and Application*. Cambridge, MA: MIT Press. ISBN: 9780262331708.

Kuo, H. (2022). "Entropy of weighted graph by edge weights by using Sombor reduced index." *Journal of Corrosion and Materials*: 47:20-28. https://doi.org/10.61336/jcm2022-3.

Li, H., *et al* (2025). "A Bayesian analysis framework for decision making with interval pairwise comparison judgements." Decision Analysis: 22(4)235-254. https://doi.org/10.1287/deca.2024.0207.

Li, J., *et al* (2025). "A method for optimizing different geometric shields of D-T neutron generators by combining BP neural network and Analytic Hierarchy Process." *Computer Physics Communications*: 307:109397. https://doi.org/10.1016/j.cpc.2024.109397.

Lindley, D. V., Tversky, A., Brown, R. V. (1979): "On the Reconciliation of Probability Assessments." *Journal of the Royal Statistical Society. Series A (General)*: 142(2)146-180. https://doi.org/10.2307/2345078.

Linkov, I., *et al* (2006). "Multicriteria decision analysis: A comprehensive decision approach of management of contaminated sediments." *Risk Analysis*: 26(1):61-78: https://doi.org/10.1111/j.1539-6924.2006.00713.x.

Liu, S., *et al* (2022). "Relative entropy of distance distribution based similarity measure of nodes in weighted graph data." *Entropy*: 24:1154. https://doi.org/10.3390/e24081154.

Lukinskiy, Valery, Lukinskly, Vladislav, Bazhina, D. (2025). "Refining the analytic hierarchy process: A statistical and methodological exploration of expert judgments." *Alexandria Engineering Journal*: 125:526-536. https://doi.org/10.1016/j.aej.2025.04.022.

Martin, R. L., & Osberg, S. (2007). Social entrepreneurship: The case for definition. *Stanford Social Innovation Review*, 2007 (Spring). https://www.ngobiz.org/picture/File/Social%20Enterpeuneur-The%20Case%20of%20Definition.pdf.

Merrick, J. R. W., van Dorp, J. R., Singh, A. (2005). "Analysis of correlated expert judgements from extended pairwise comparisions." *Decision Analysis*: 2(1):1-65. https://doi.org/10.1287/deca.1050.0031.

Moon, J. W. (2015). *Topics on Tournaments in graph Theory*. Dover. ISBN: 9780486804736. (Originally, New York: Holt, Rinehart, and Winston 1968).

Mowshowitz, A. & Dehmer, M. (2012). "Entropy and the complexity of graph revisited." *Entropy*: 14:559-570. https://doi.org/10.3390/e14030559.

Nishizawa, K. (1999). "A method to improve consistency of the comparison matrix based on iteration." *Proceedings of the International Symposium on the Analytic Hierarchy Process* (ISAHP 1999), Kobe, Japan, August 12–14, 1999, pp 89-95. https://doi.org/10.13033/isahp.y1999.004.

Olive, D. J. (2017). *Linear Regression*. Cham, CH: Springer Nature. https://doi.org/10.1007/978-3-319-55252-1.

Pant, S. et *al* (2022). "Consistency Indices in Analytic Hierarchy Process: A Review." *MDPI Mathematics*: 10(8):1206. https://doi.org/10.3390/math10081206.

Pearl, J. (1988). *Probablistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Elsevier. ISBN: 9780080514895.

Pettigrew, R. (2016). *Accuracy and the Laws of Credence*. Oxford, UK: Oxford University Press. ISBN: 9780198732716.

Ren, Q. & Sun, M. (2025). "Using AHP-Entropy method to explore the influencing factors of spatial demand of EVs public charging stations: A case study of Jinan, China." *Journal of Cleaner Production*: 491:144779. https://doi.org/10.1016/j.jclepro.2025.144779.

Rindova, V. P. & Martins, L. L. (2024). "The imagination advantage: Why and how strategists combine knowledge and imagination in developing theories." Strategy Science: 9(4): 297-314. https://doi.org/10.1287/stsc.2024.0184.

Rosiak, D. (2022). *Sheaf Theory through Examples*. Cambridge, MA: The MIT Press. ISBN: 9780262362375. https://doi.org/10.1287/inte.24.6.19.

Saaty, T. L. (1986). "Axiomatic foundation of the Analytic Hierarchy Process." *Management Science*: 32(7)841-855. https://doi.org/10.1287/mnsc.32.7.841.

Saaty, T. L. (1987). "Risk — Its priority and probability: The Analytic Hierarchy Process." *Risk Analysis*: 7(2):159-172. https://doi.org/10.1111/j.1539-6924.1987.tb00980.x.

Saaty, T. L. (1994). "How to make a decision: The Analytic Hierarchy Process." *Applied Analytics*: (24)6:19-43. https://doi.org/10.1287/inte.24.6.19.

Saaty, T. L. (2003). “Decision-making with the AHP: Why is the principal eigenvector necessary.” *European Journal of Operational Research*: 145(1):85-91. https://doi.org/10.1016/S0377-2217(02)00227-8.

Saaty, T. L. & Vargas, L. G. (2013). *Decision Making with the Analytic Network Process*. New York: Springer. ISBN: 9781461472797.

Sáenz-Royo, C. & Chiclana, F. (2026). “Contribution of the ratio scale of expert judgments in the analytic hierarchy process.” *Expert Systems with Applications*: 296(C):128938. https://doi.org/10.1016/j.eswa.2025.128938.

Saliconi, S. & Prioli, M (2018). *Measuring Uncertainty within the Theory of Evidence*. Cham, CH: Springer Nature. ISBN: 9783319741390.

Savage, L. (1972). *The Foundations of Statistics*. Garden City, NY: Dover. ISBN: 9780486137100. (Originally Savage, L. (1954). *The Foundations of Statistics*. New York: Wiley.)

Shafer, G. (1976). *A Mathematical Theory of Evidence*. Princeton, NJ: Princeton University Press. ISBN: 9780691100425.

Shen, Z. & Wu, K. (2023). “AHP-entropy Weight Method Based Comprehensive Evaluation Approach for Distribution Network Research.” 2023 3rd International Conference on New Energy and Power Engineering (ICNEPE), November 24-26, 2023, Huzhou, China. https://doi.org/10.1109/ICNEPE60694.2023.10429261.

Shenoy, P. P. (2023). “Making inferences in incomplete Bayesian networks: A Dempster-Shafer belief-function approach.” *International Journal of Approximate Reasoning:* 160(2023):108967. https://doi.org/10.1016/j.ijar.2023.108967.

Society of Decision Professionals. “Decision Quality”. https://www.decisionprofessionals.com/who-we-are/decision-quality. Accessed May 27, 2026.

Spetzler, C. S & Staël Von Holstein, C. S. (1975). “Exceptional Paper—Probability Encoding in Decision Analysis.” Management Science: 22(3):340-358. https://doi.org/10.1287/mnsc.22.3.340.

Spetzler, C., Winter, H., Meyer, J. (2016). *Decision Quality Value Creation from Better Business Decisions*. New York: Wiley. ISBN: 9781119176657. https://doi.org/10.1016/j.jlp.2019.104024.

Sun, G., Kroesen, M. Rezael, J. (2025). *Decision Analysis*: 22(4):284-304. “ https://doi.org/10.1287/deca.2024.0308.

Swanson, E. B. (2024). “Technology entrepreneurship is more than one might think.” *Information and Organization*: 34:100512. https://doi.org/10.1016/j.infoandorg.2024.100512.

Syed, Z., Shabarchin, O., Lawryshyn, Y. (2020). “A novel tool for Bayesian reliability analysis using AHP as a framework for prior elicitation.” *Journal of Loss Prevention in the Process Industries*: 64:104024. https://doi.org/10.1016/j.jlp.2019.104024.

Townsend, D. M., Hunt, R. A., Rady, J. (2024). “Chance, probability, and uncertainty at the edge of human reasoning: What is Knightian uncertainty?” *Strategic Entrepreneurship Journal*: 18(3)451-474. https://doi.org/10.1002/sej.1516.

Townsend, D. M., *et al* (2023). “Are futures computable? Knightian uncertainty and artificial intelligence.” *Academy of Management Review*. (50)2:415:440. https://doi.org/10.5465/amr.2022.0237.

Vaidya, O. S. & Kumar S. (2006). “Analytical hierarchy process: An overview of applications.” European Journal of Operational Research: 169(1):1-29. https://doi.org/10.1016/j.ejor.2004.04.028.

Virtanen, P., *et al* (2020). “SciPy 1.0: fundamental algorithms for scientific computing in Python”. *Nature Methods*: 17:261-272. https://doi.org/10.1038/s41592-019-0686-2.

von Hayek, Fredrick (1974), “The pretence of knowledge”, Nobel Prize Lecture, December 11, 1974, https://www.nobelprize.org/prizes/economic-sciences/1974/hayek/lecture/.

Wong, A. K. C. & You, M. (1985). “Entropy and distance of random graph with aplication to structural pattern recognition.” *IEEE Transactions on Pattern Analysis and Machine Intelligence*: PAMI-7(5):599-609. https://doi.org/10.1109/TPAMI.1985.4767707.

Xiao, H., *et al* (2025). “A simulation optimization approach for weight valuation in analytic hierarchy process.” *European Journal of Operations Research*: 321(3):851-864. https://doi.org/10.1016/j.ejor.2024.10.018.

Zellweger, T., & Zenger, T. (2023). “Entrepreneurs as scientists: A pragmatist’s approach to producing value out of uncertainty.” Academy of Management Review: 48(3):397-408. https://doi.org/10.5465/amr.2020.0503.

Zeng, D., *et al* (2025). “Material selection of titanium alloy pipelines considering multi-criteria in an acidic environment based on analytic hierarchy process.”

*International Journal of Pressure Vessels and Piping*: 2016:105461. https://doi.org/10.1016/j.ijpvp.2025.105461.